

4.9.3 The Internet

4.9.3.1 The Internet and how it works

Content	Additional information
Understand the structure of the Internet.	
Understand the role of packet switching and routers.	
Know the main components of a packet.	
Define: • router • gateway. Consider where and why they are used.	
Explain how routing is achieved across the Internet.	
Describe the term 'uniform resource locator' (URL) in the context of internetworking.	
Explain the terms 'fully qualified domain name' (FQDN), 'domain name' and 'IP address'.	
Describe how domain names are organised.	
Understand the purpose and function of the domain service and its reliance on the Domain Name Server (DNS) system.	
Explain the service provided by Internet registries and why they are needed.	

4.9.3.2 Internet security

Content	Additional information
Understand how a firewall works (packet filtering, proxy server, stateful inspection).	
Explain symmetric and asymmetric (private/public key) encryption and key exchange.	
Explain how digital certificates and digital signatures are obtained and used.	

Content	Additional information
Discuss worms, trojans and viruses, and the vulnerabilities that they exploit.	
Discuss how improved code quality, monitoring and protection can be used to address worms, trojans and viruses.	